

Dermabiotics by Inner-LAB

Probiotics Technology Platform for Skin Care



Probiotics

The WHO(World Health Organization)/ FAO(Food and Agriculture Organization of the UN) define probiotics as “ **Live micro-organisms**, which when administered in adequate amounts, confer a health benefit on the host” (WHO/FAO, 2002)

- Individual diversity of skin microbiota due to stress or different life styles
- Skin turnover makes difficult to improve skin condition by simply applying probiotics on skin surface

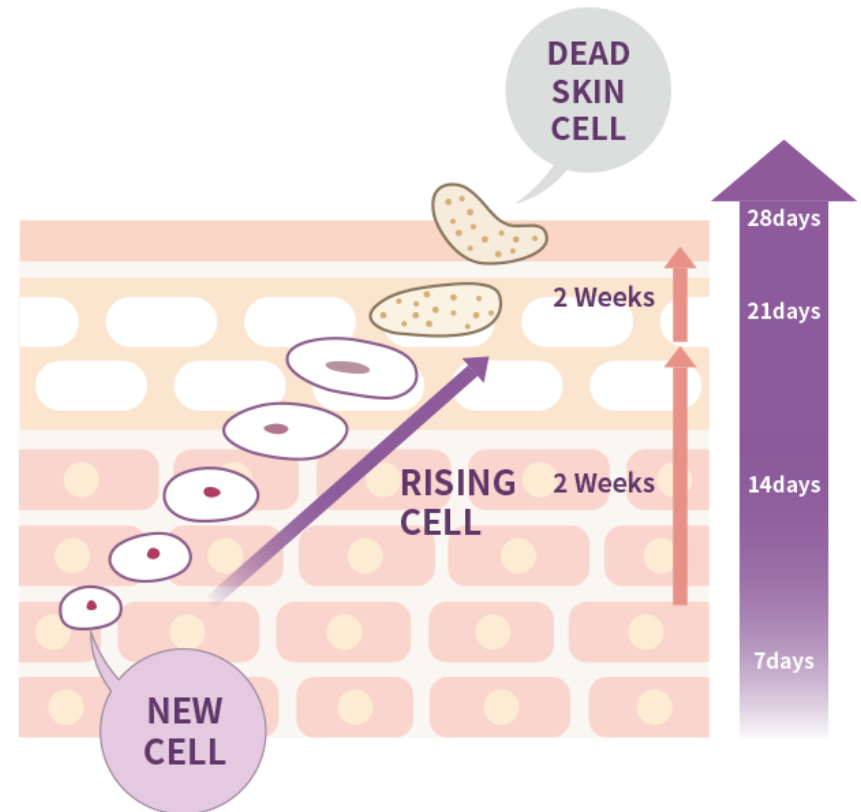
Behavioral Factors

Stressful Life & Different Hygiene Habit



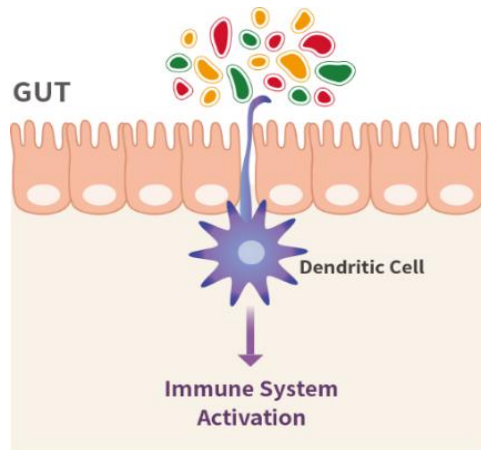
Skin Cells Turnover

Removal of Skin Microbiome by Skin Turnover



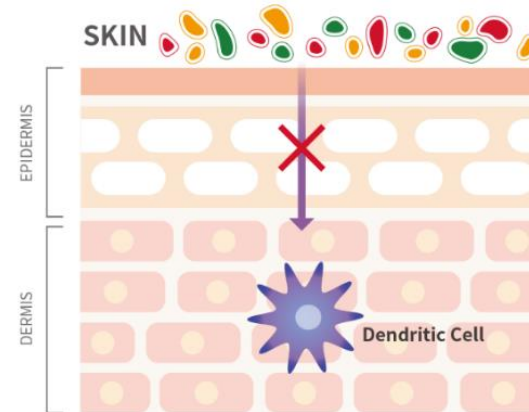
- Gut dendritic cells are present outside the intestinal mucosa
- Skin dendritic cells are present inside the skin

Different Mechanisms of Gut & Skin



[GUT]

- Gut dendritic cells (DCs) are present outside the **intestinal mucosa**
- Probiotics directly induce the activation of intestinal immune system



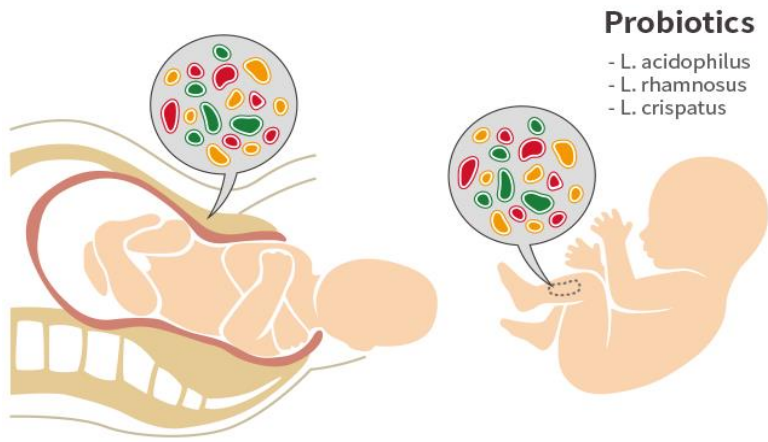
[SKIN]

Skin dendritic cells (DCs) are located inside the skin

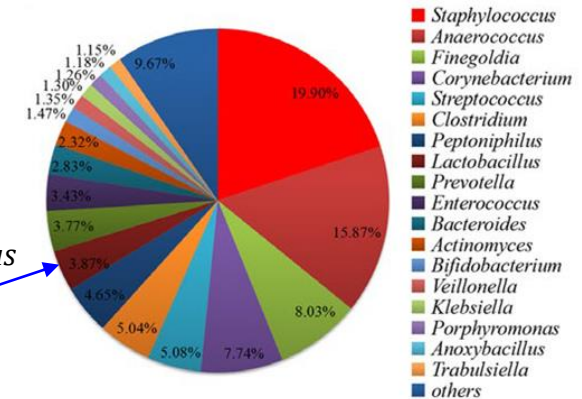
- ① Live microorganism signal towards immune system (DCs) is weak due to the restriction in skin penetration
 - ② Applying live microorganisms over skin can cause reverse effects; antagonism (against live microorganism) & skin irritation (by lactic acid)
- It is required to develop **a new bio-cosmeceutical material** that not only can **minimize skin trouble** but also can get **the active probiotic substances to arrive at skin dermis safe and sound**.

- Probiotics are important in first formation of new born babies' microbiome at birth, connected to immune system.
- Birth Canal Seeding is a primitive form of cosmetics.

Initial Formation of Skin Barrier



Lactobacillus rhamnosus
Lactobacillus acidophilus
Lactobacillus casei
Lactobacillus gasseri
Lactobacillus crispatus
Lactobacillus paracasei



Birth Canal Microbiome



Infant's Skin Microbiome

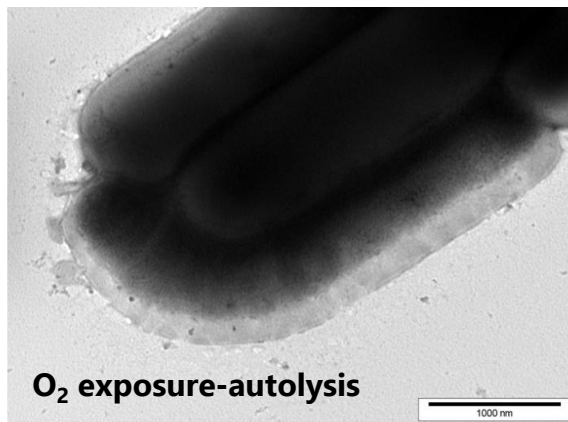
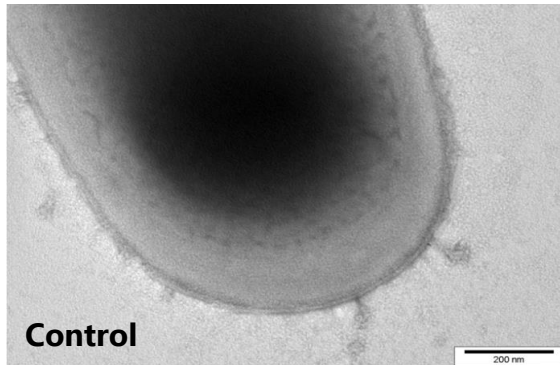
Cold Spring Harb Perspect Med., 2015

Birth Canal Seeding for babies born by caesarean section (C-section)

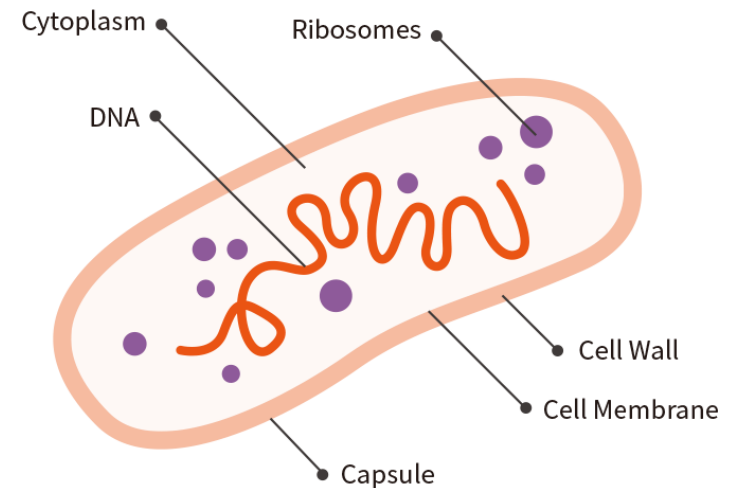


- Probiotics are good at activating immune and metabolism cells
- BUT, the effects of Probiotics are small due to skin structure
- Thus, we need to improve skin penetrance and reduce antagonism → **SKB's Dermabiotics!**

Lactobacillus Autolysis



SKB's Focus



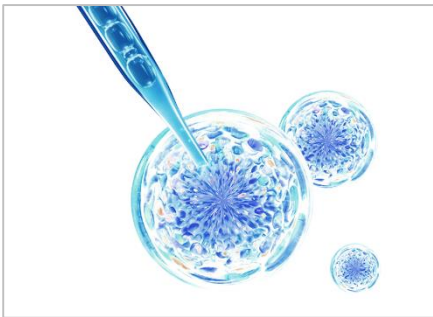
[Cytosol in Probiotics]

- **Basic components of probiotics** such as RNA/DNA, except cell organelles such as mitochondria and endoplasmic reticulum, and **essential to cell growth**
- A translucent fluid which contains low molecular weight of **bioactive materials such as nucleic acid (< 2KDa)** that can penetrate into the skin

- Utilizing bio-active substances in probiotics cell for skin care ingredient
- Presenting skin microbiome solution through inner skin care

Probiotics Inner Skin Solution ***that cares skin from inside***

Make your skin healthy with low molecular weight of probiotic 'cell free' lysate



- ① SKB owns probiotics strains in its cell bank which are involved in the early stage of immunity formation.
- ② They are registered at "U.S. GRAS" (*Generally Recognized As Safe*)

DERMABIOTICS STRAINS OF SKB'S OWN

No	Strain Name	Strain No.	Main Application
1	<i>Lactobacillus acidophilus</i>	1014	Probiotic cosmetic/ Skin Microbiome
2	<i>Lactobacillus delbrueckii</i> subsp. <i>bulgaricus</i>	1086	Probiotic cosmetic/ Skin Microbiome
3	<i>Lactobacillus fermentum</i>	1097	Probiotic cosmetic/ Skin Microbiome
4	<i>Lactobacillus gasseri</i>	1105	Probiotic cosmetic/ Skin Microbiome
5	<i>Lactobacillus paracasei</i>	1196	Probiotic cosmetic/ Skin Microbiome
6	<i>Lactobacillus plantarum</i>	1303	Probiotic cosmetic/ Skin Microbiome
7	<i>Lactobacillus reuteri</i>	1242	Probiotic cosmetic/ Skin Microbiome
8	<i>Lactobacillus rhamnosus</i>	1259	Probiotic cosmetic/ Skin Microbiome
9	<i>Lactococcus lactis</i> subsp. <i>hordniae</i>	2033	Probiotic cosmetic/ Skin Microbiome
10	<i>Bifidobacterium longum</i> subsp. <i>longum</i>	7072	Probiotic cosmetic/ Skin Microbiome
11	<i>Saccharomyces cerevisiae</i>	9015	Yeast cosmetic/ Skin Microbiome

- Removes probiotic cell walls and eliminates the possibility of antagonist
- Penetrates active ingredients that activates the immune signaling in dermis and improves skin flora and skin health

2-Steps Fermentation

Culture of LAB(10^{10} CFU/ml)



Membrane Filtration/Remove Lactic acid



Di Cerbo et al. BMC Microbiology 2013, 13:243

Post-fermentation with HA



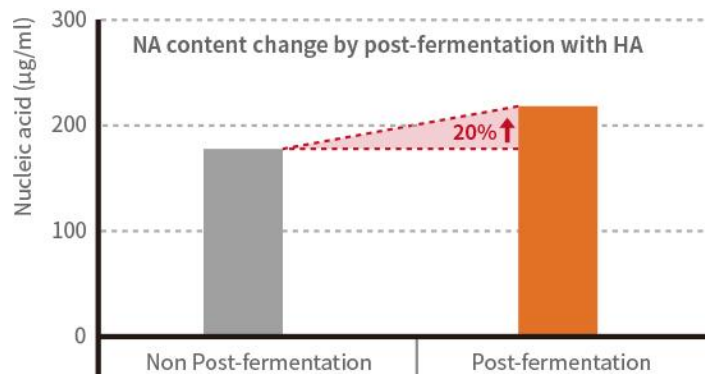
High Pressure Extraction



Probiotic Lysate Filtrate



Skin Care by Penetration of Dermabiotics

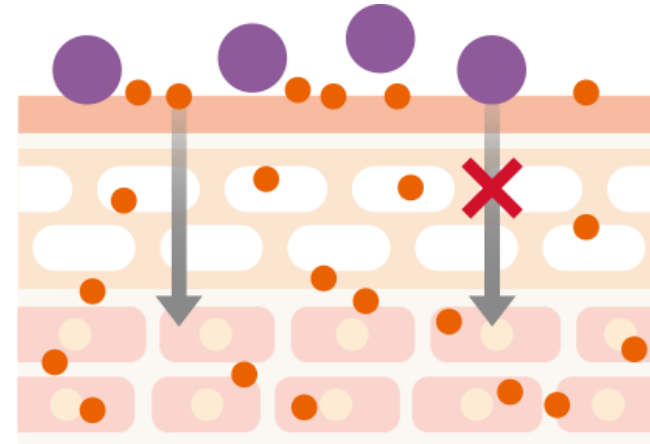


Skin Penetration

❖ Trojan Horse Motive ; Skin penetration system

Probiotic Lysate Filtrate (Dermabiotics)

Dermabiotics : 2 KDa↓



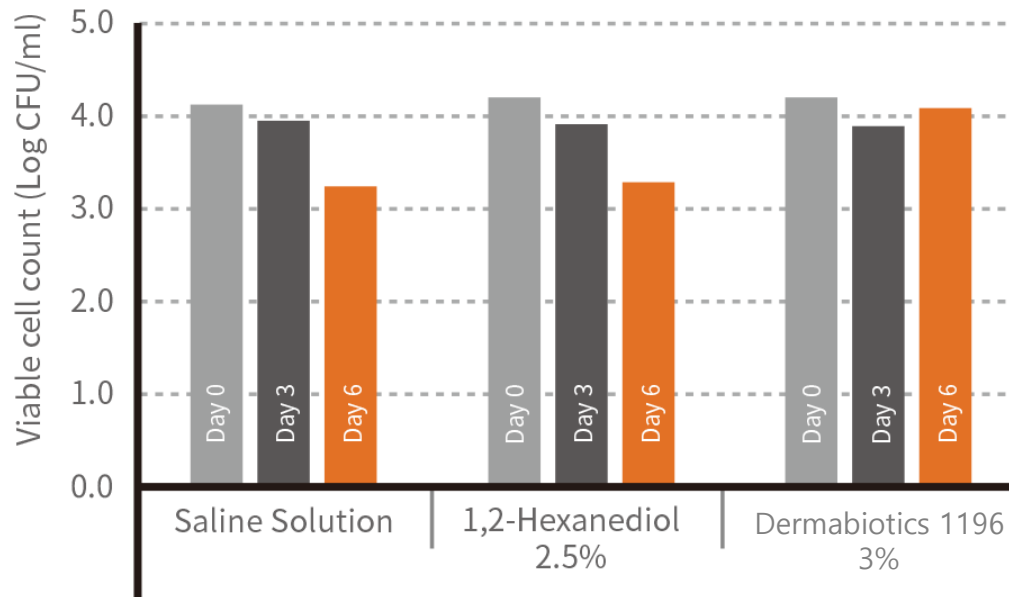
- 10 billion microorganism culture
- Minimizes skin irritation by Lactic Acid, and increases cytosol contents by post-fermentation with HA

- High pressure extraction makes it easy to obtain cell-free probiotic lysates; Cytosol that can penetrate skin while not recognized as antigens to induce antagonist

-Dermabiotics SKB1196 maintains the number of viable cells of *Staphylococcus epidermis* that is one of skin flora, in contrast to the other solvents.

**Lactobacillus paracasei* SKB1196

Pre-Test for Skin Microbiome Feasibility (in-vitro)



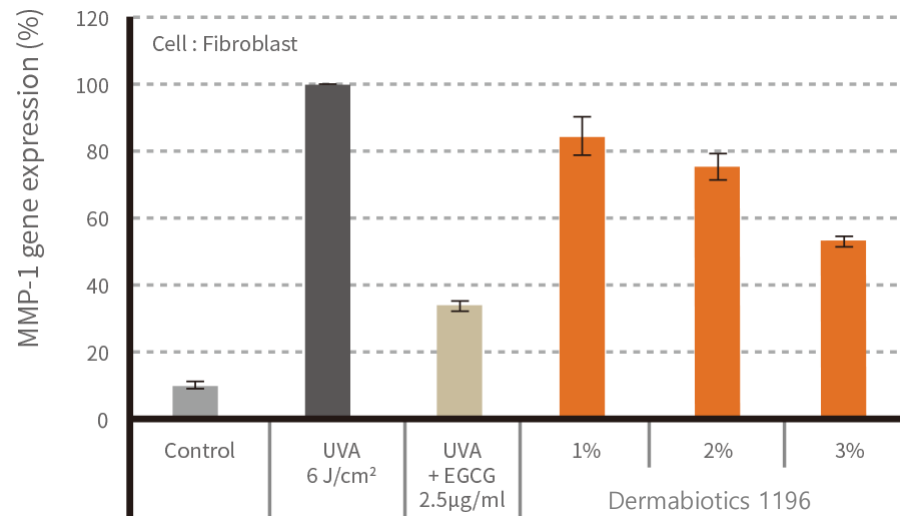
[*Staphylococcus epidermis* impact assessment]

"Dermabiotics SKB 1196" was applied onto *Staphylococcus epidermis*, skin flora, and showed it devotes to maintaining and increasing the cell viability.

- Dermabiotics SKB1196 : inhibits the expression of MMP-1 and upregulates the expression of collagen synthesis

**Lactobacillus paracasei* SKB1196

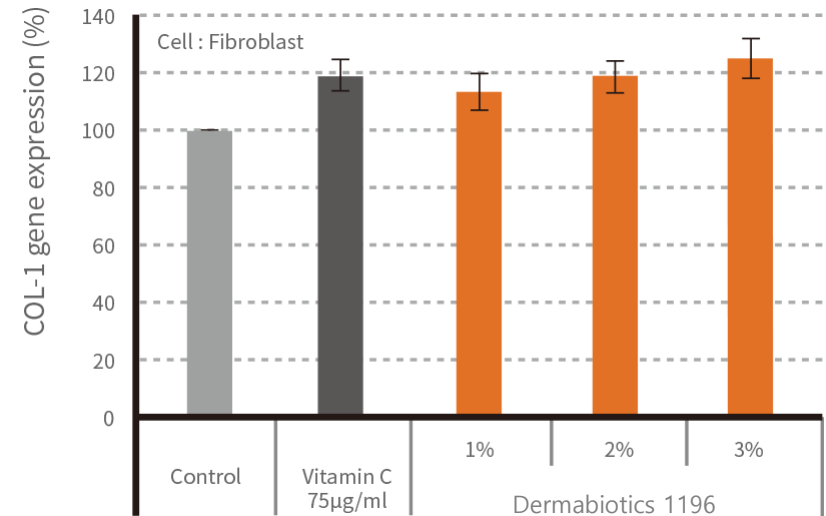
Anti-wrinkle (in-vitro)



[MMP-1]

- ❑ Dermabiotics strengthens anti-wrinkle effect by mRNA expression of MMP-1

→ It helps for skin elasticity as reducing the decomposition of collagen



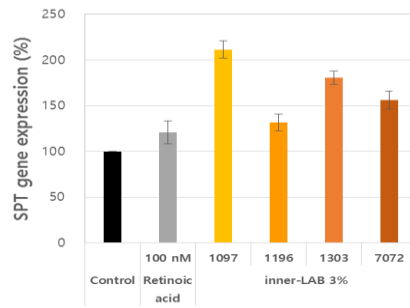
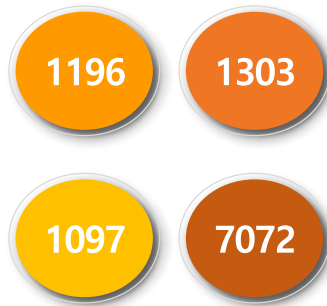
[Collagen type 1]

- ❑ Dermabiotics enhances anti-wrinkle effect by mRNA expression of Collagen type 1

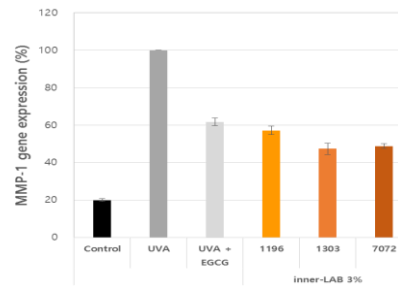
→ Biosynthesis of collagen maintains skin elasticity and prevents skin wrinkle

- Probiotic strains are classified by skin efficacy through in-vitro evaluation

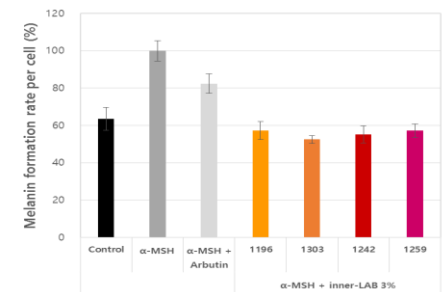
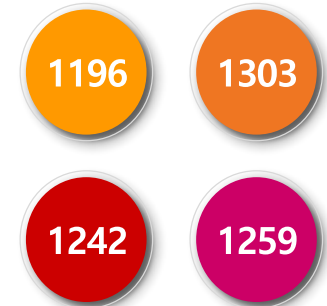
Skin Barrier



Anti-wrinkle



Brightening



- **Marker of DERMABIOTICS** : Nucleic acid 200 µg/ ml ↑
- **ISO16128** : Natural Index 1.0

**Lactobacillus paracasei SKB1196*

Dermabiotics SKB 1196

SPECIFICATION

ANALYTICAL TESTS	SPECIFICATIONS
INCI Name	GLYCERINE LACTOBACILLUS FERMENT LYSATE SODIUM HYALURONATE
Description	Light yellow liquid
pH(Neat)	5.5 ~ 6.5
Specific Gravity (d20/20)	0.990 ~ 1.020
Heavy Metals	≤ 20ppm
Arsenic	≤ 2ppm
Preservative	None
Nucleic acid	≥ 200 µg /ml
Microbiological Purity	
- Total bacterial count	≤ 100 CFU/ml
- Yeast & molds	≤ 10 CFU/ml

PATENT

Patent No.	10-2018-0126226
Applicant	SK BIOLAND
Patent	Method for preparing nucleotides of lactic acid bacteria inclusion-complexed with low-molecular functional biomaterial and Cubisome prepared thereby

Lava-Dermabiotics by Inner-LAB

Probiotics Technology Platform for Skin Care

- SKB have developed another probiotics cultivated with Jeju Lava Sea Water which is rich in natural minerals

Lava Sea-Water Probiotics Solution

Strengthen skin barrier with rich mineral & lysate of probiotics cultivated with Lava Sea Water



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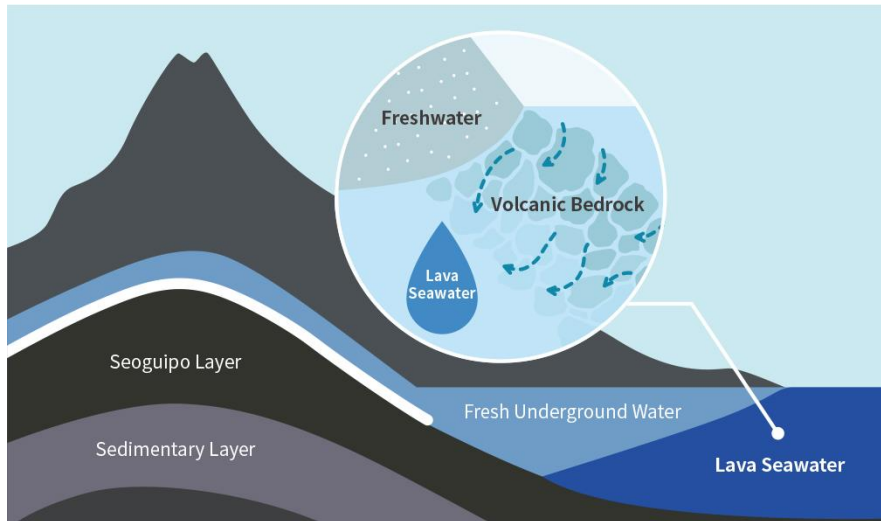
***Lava Sea Water rich in
minerals such as Magnesium
and Selenium***

***Probiotics Lysate
strengthening skin barrier***

***Lava Sea Water
Probiotics for Skin Care***

- Natural resourced filtered by Jeju basalts & sustainable supply by SKB Jeju plant ,
- Rich minerals with natural-derived safety and homeostasis

Feature



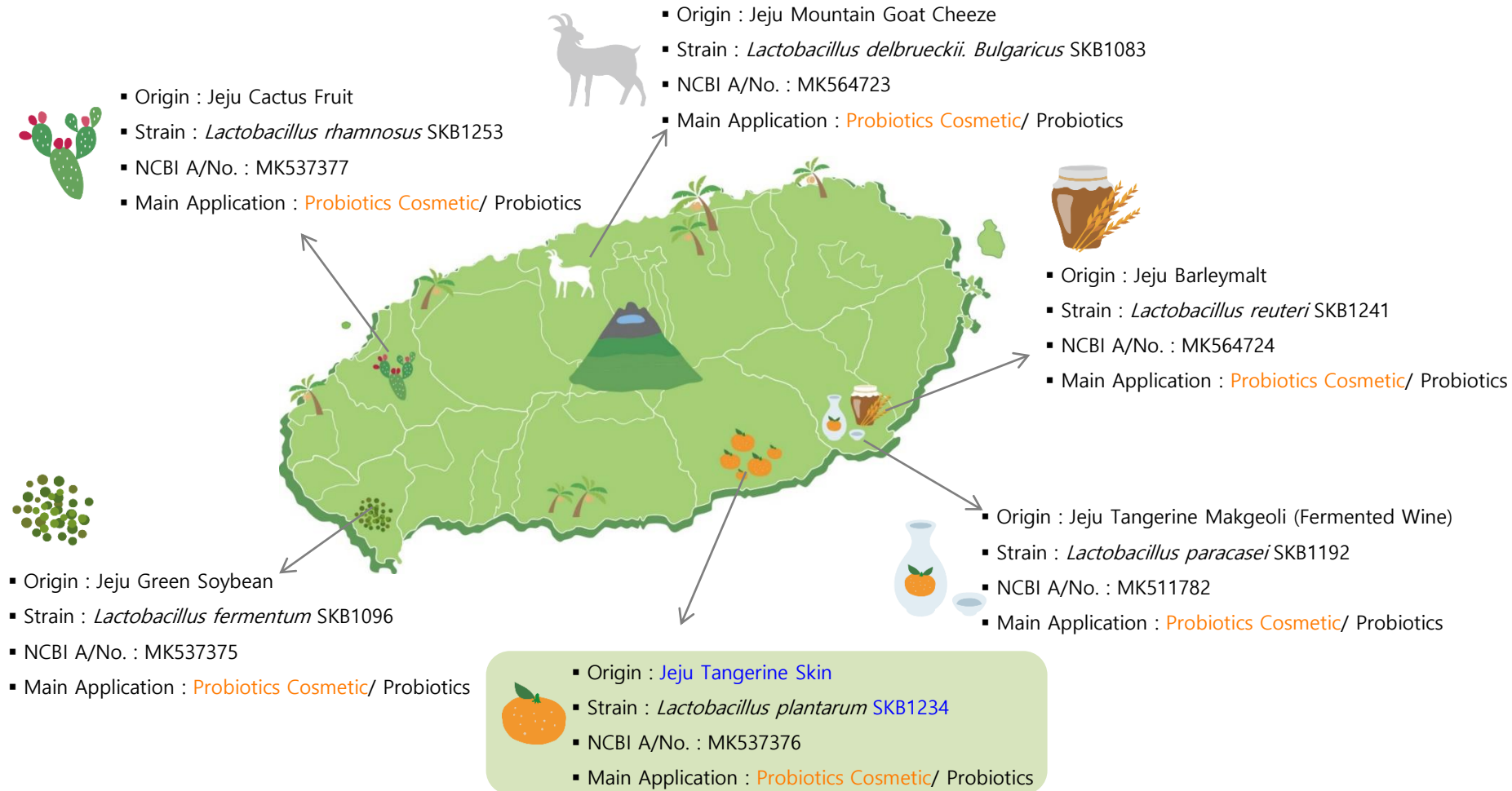
- Clean and Pure Sea Water filtered by underlying basalts
- Stable Sea Water with constant levels of pH, salts and etc
- High-value natural resources for the use of food and cosmetic
- Safe natural resources free from harmful substances

Comparison of natural minerals composition

Components	Concentration (mg/mL)		
	Sea water	Deep ocean water	Jeju Lava Sea Water
Na	10,762	11,000	10,510
Mg	1,294	1,270	1,295
Ca	413	474	399
K	387	403	400
Br	67	80.3	63.4
Sr	7.8	8.03	7.7
Zn	0.00035	<0.005	0.019
Fe	0.00003	<0.03	0.015
Mn	0.00002	<0.005	0.008
V	0.002	-	0.015
Se	0.0001	<0.005	0.013
Ge	-	-	0.002

- SK bioland owns another probiotic strains isolated from Jeju
- They are registered at U.S. NCBI (The National Center for Biotechnology Information)

SKB's Jeju Heritage Probiotics



- Boosts mineral & nucleic acid contents in cell by cultivating with Lava Sea Water
- Removes irritative probiotics metabolites / Extracts lava sea water probiotics lysate by high pressure extraction

2-Steps Fermentation

Lava Sea-Water Probiotics Solution

**Lactobacillus plantarum* SKB1234 Jeju Tangerine

Lava Sea-Water Culture of LAB(10^{10} CFU/ml)



Membrane Filtration / Remove Lactic acid



**Post-fermentation with
Lava Sea-Mineral Water**



High Pressure Extraction

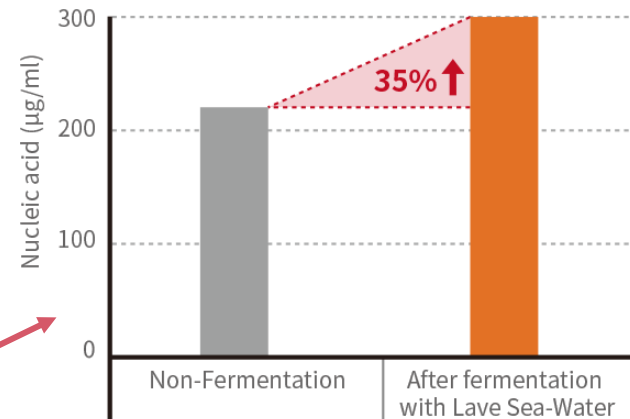


Probiotic Lysate Filtrate

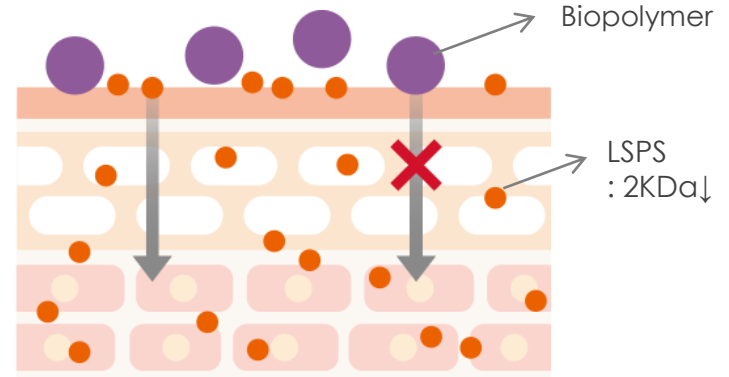


Skin Care by Penetration of Lava-Dermabiotics

Change of Nucleic Acid content after fermentation with lava sea water



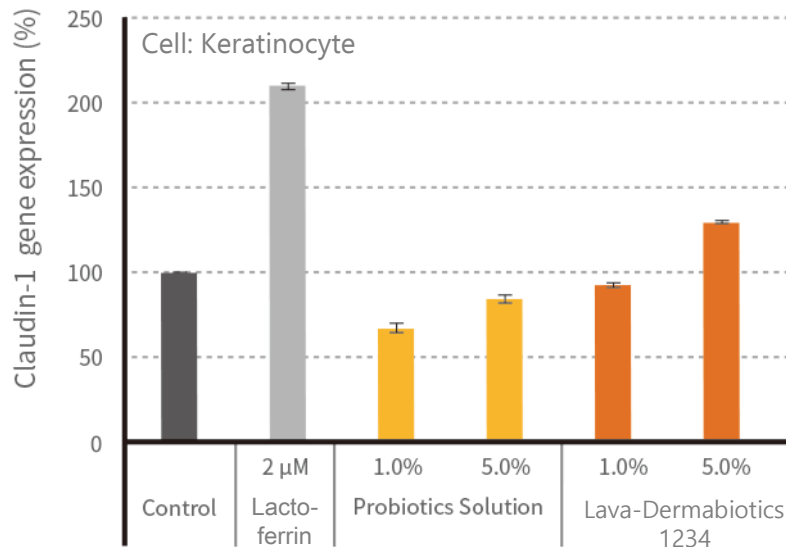
Lava Sea-Water Probiotics Solution



- Lava-Dermabiotics SKB1234 : Probiotics Lysate & Lava Sea Water create a synergistic effect on skin barrier strengthening

**Lactobacillus plantarum* SKB1234 Jeju Tangerine 

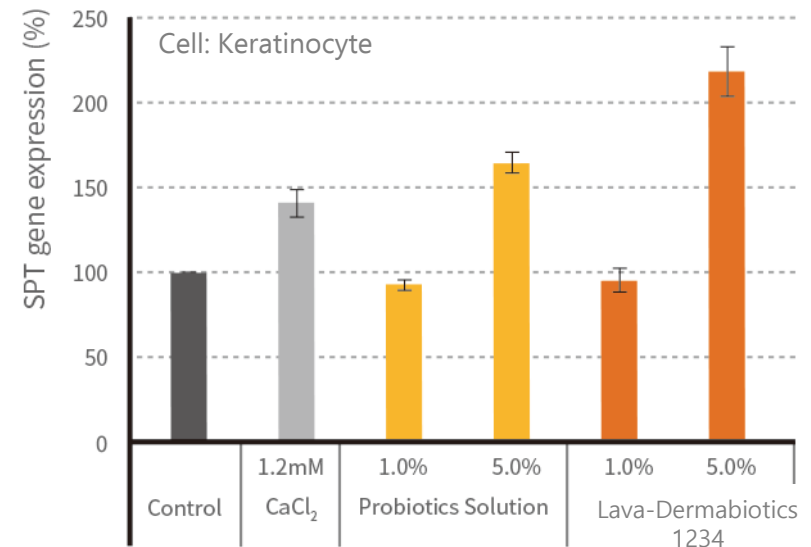
Barrier Formation (in-vitro)



[Claudin-1]

- Lava Dermabiotics strengthens skin barrier by Claudin-1 gene expression
- Claudin-1 is intercellular tight junction protein that strengthens skin barrier

Skin Moisture Barrier (in-vitro)



[Serine palmitoyltransferase]

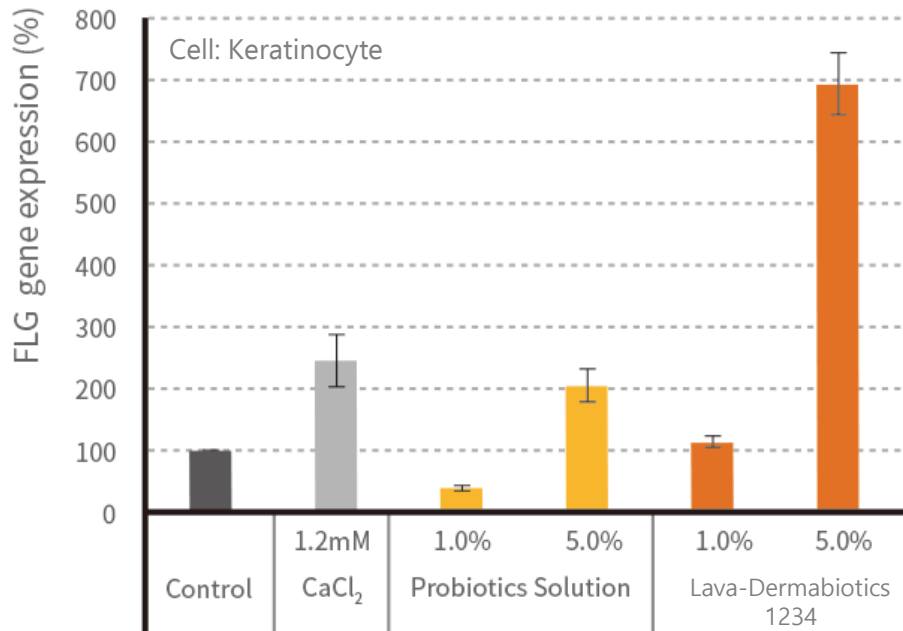
- Lava Dermabiotics strengthens skin moisture barrier by stimulating SPT gene expression
- SPT is an enzyme related to ceramide synthesis

- Lava-Dermabiotics SKB1234 : Probiotics Lysate & Lava Sea Water create a synergy effect on skin barrier strengthening

**Lactobacillus plantarum* SKB1234 Jeju Tangerine



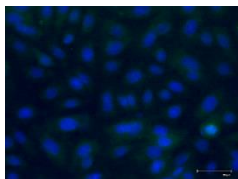
Skin Moisture Barrier (in-vitro)



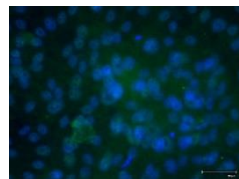
[Filaggrin]

Lava Dermabiotics strengthens skin moisture barrier by Filaggrin gene expression

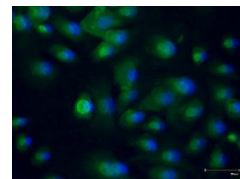
→ Filaggrin helps to generate strong skin barrier by creating stratum corneum from epidermal proliferation and differentiation



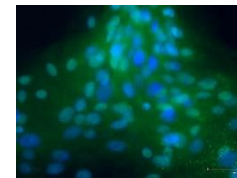
Control 5.0%



CaCl₂ 1.2 mM



Probiotics Solution 5.0%



Lava-Dermabiotics solution 5.0%

Cell proliferating!

- Marker of LAVA DERMABIOTICS : Nucleic acid 300µg/ ml ↑
- ISO16128 : Natural Index 1.0

Lava Dermabiotics SKB 1234

*Lactobacillus plantarum SKB1234 Jeju Tangerine



SPECIFICATION

ANALYTICAL TESTS	SPECIFICATIONS
INCI Name	GLYCERINE
(Chinese Compliant)	LACTOBACILLUS FERMENT LYSATE
Description	Light yellow liquid
pH(Neat)	5.0 ~ 6.5
Heavy Metals	≤ 20ppm
Arsenic	≤ 2ppm
Preservative	None
Nucleic acid	≥ 300 ug/ml
Microbiological Purity	
- Total bacterial count	≤ 100 CFU/ml
- Yeast & molds	≤ 10 CFU/ml

PATENT (manufacturing process)

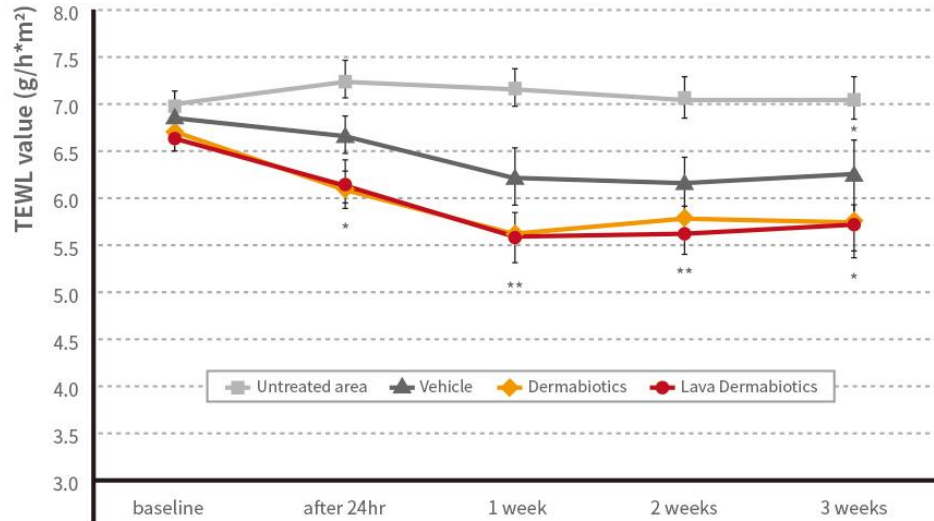
Patent No.	Registration is pending
Applicant	SK BIOLAND CO., LTD
Patent	[tentative title] Method for preparing probiotic lysate solution with jeju lava sea water minerals to enhance the skin barrier function and skin microbiome improvement

- **Test Method**

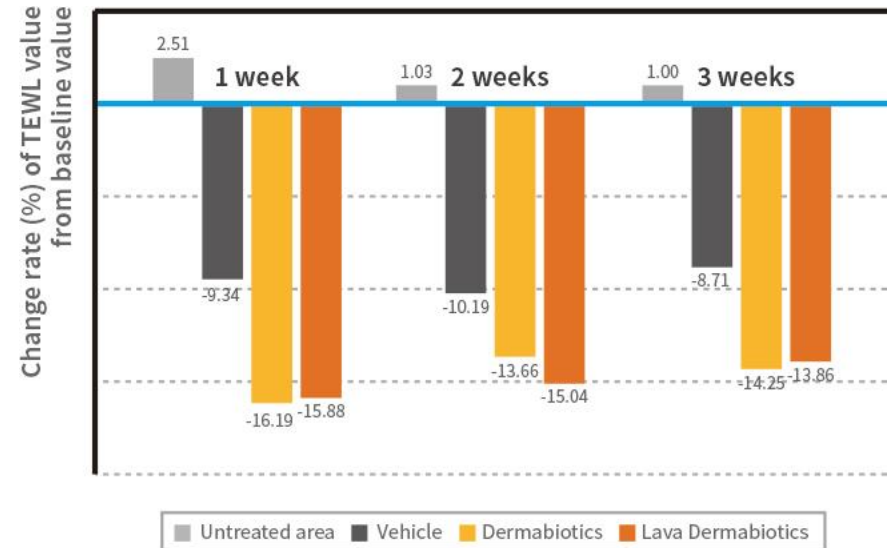
- Test Type : **Corneometer & TEWL**
- Test products : **Dermabiotics SKB1196**
Lava-Dermabiotics SKB1234
- Volunteers: 20 Females (age: 30's on average)
- Test formulation : Creams that contain the products each (3 %)
- Condition: Apply $50\mu\ell$ ($10\mu\ell/\text{cm}^2$) on the forearm (5cm^2) once a day for 3 weeks
- Period: March 30 ~ 3 weeks
- Measurement: Before application, 24 hours after application (short term)
1, 2 and 3weeks after application (long term)

- skin barrier function result (TEWL, 3 weeks)

TEWL value by the application of creams (n=20)

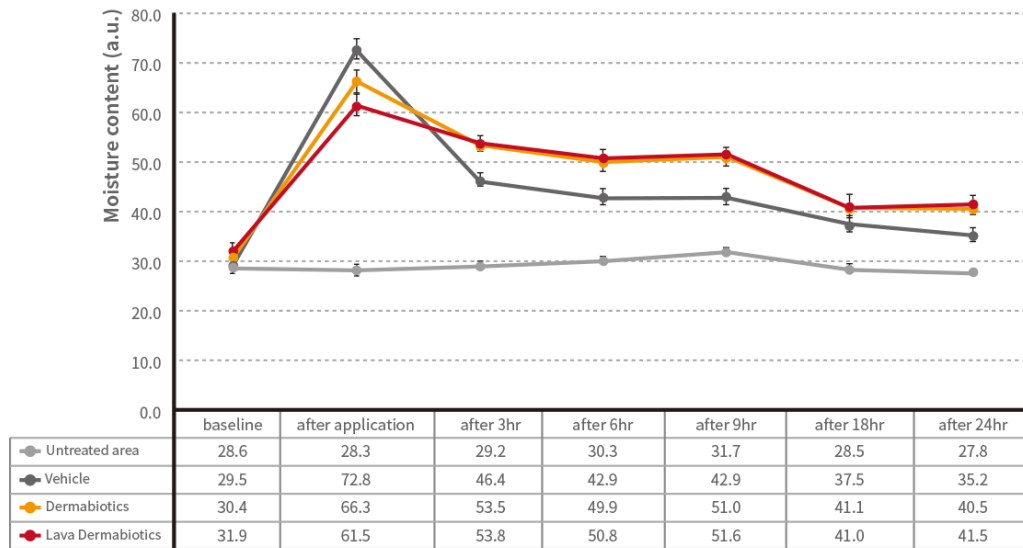


Change rate (%) of TEWL value by the application of creams (n=20)

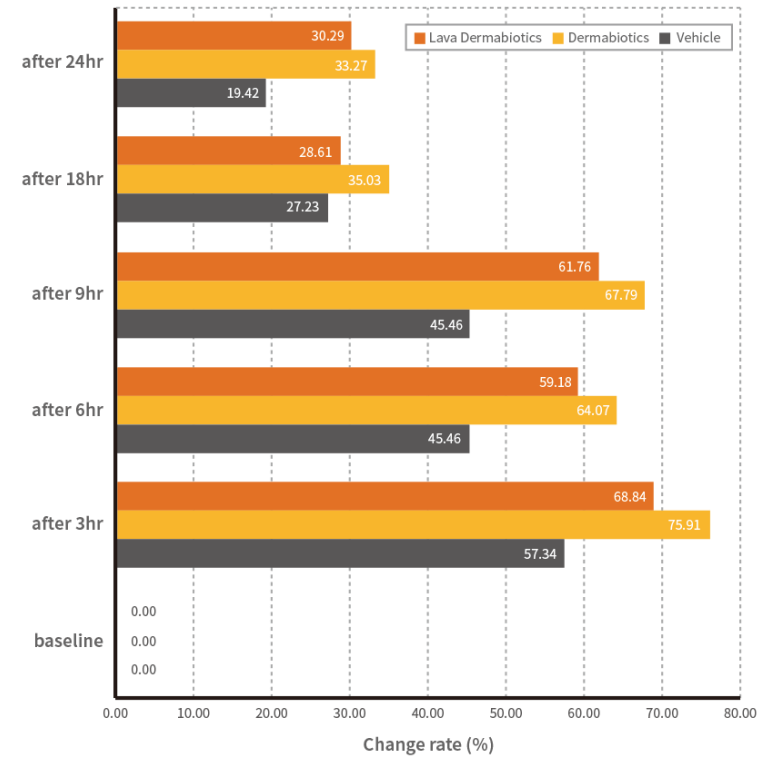


- skin hydration result (corneometer, 24 hours)

Moisture content(A.U.)
by the application of the creams (n=20)

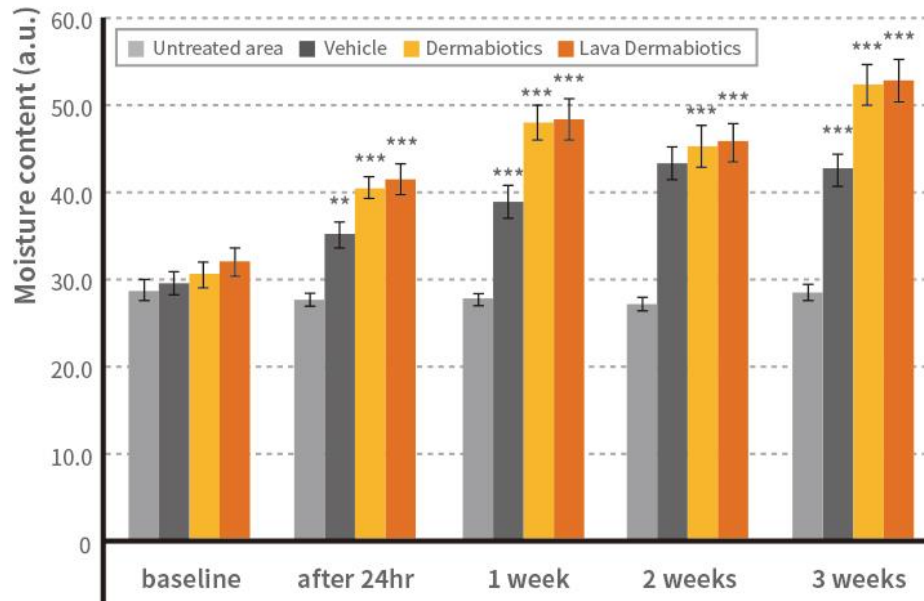


Water holding capacity (%) of moisture content
by the creams (n=20)

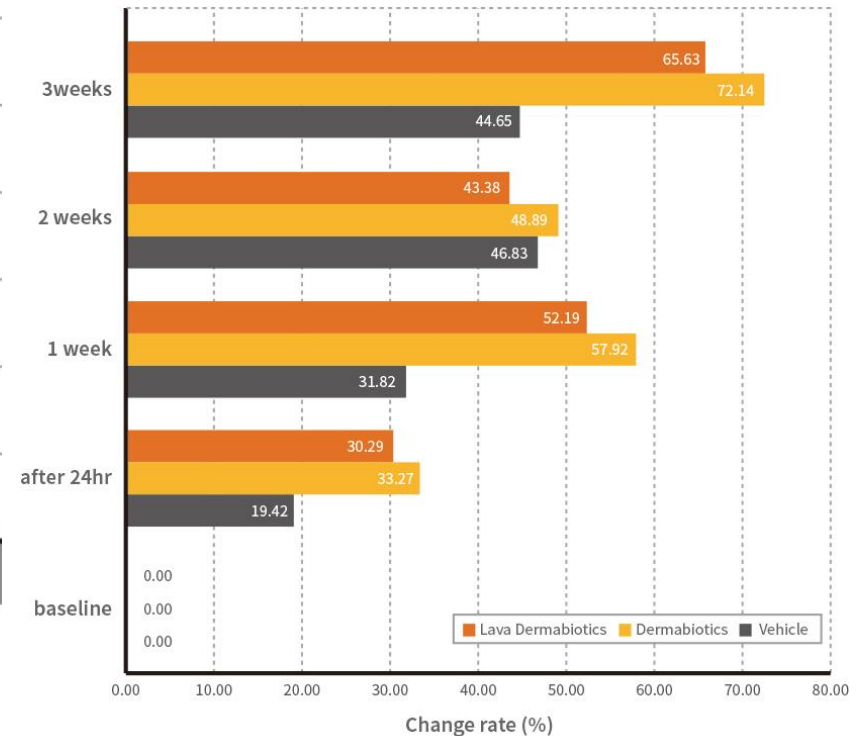


- skin hydration result (corneometer, 3 weeks)

Moisture content(A.U.)
by the application of the cream (n=20)

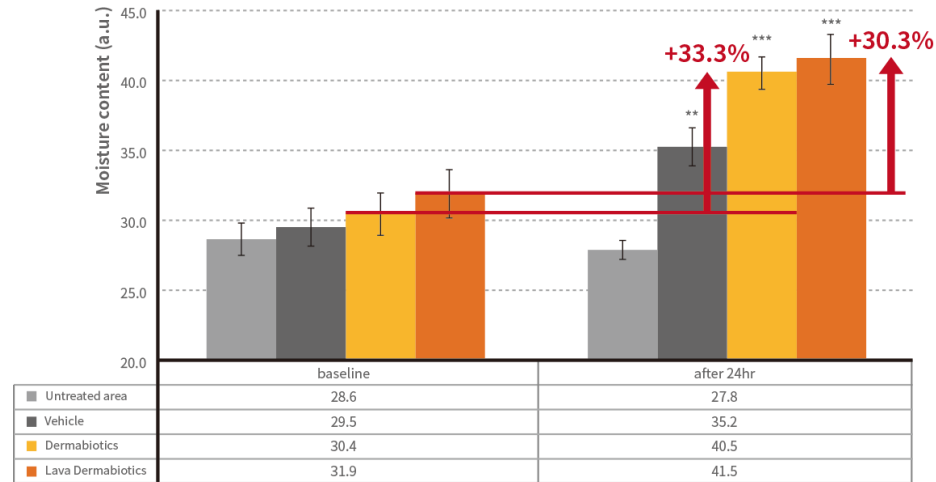


Water holding capacity (%) of moisture content
by the creams (n=20)

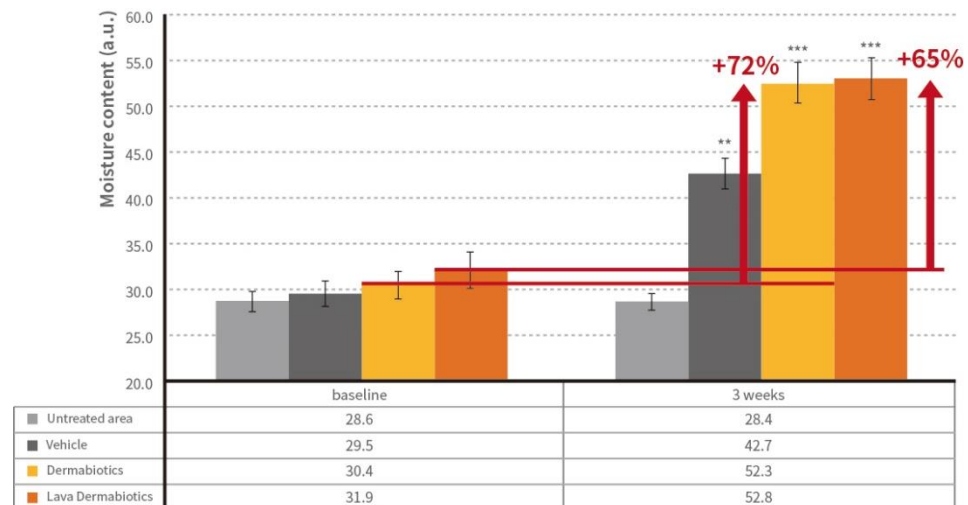


- Skin hydration result summary

- ❖ Short-term (24 hr)

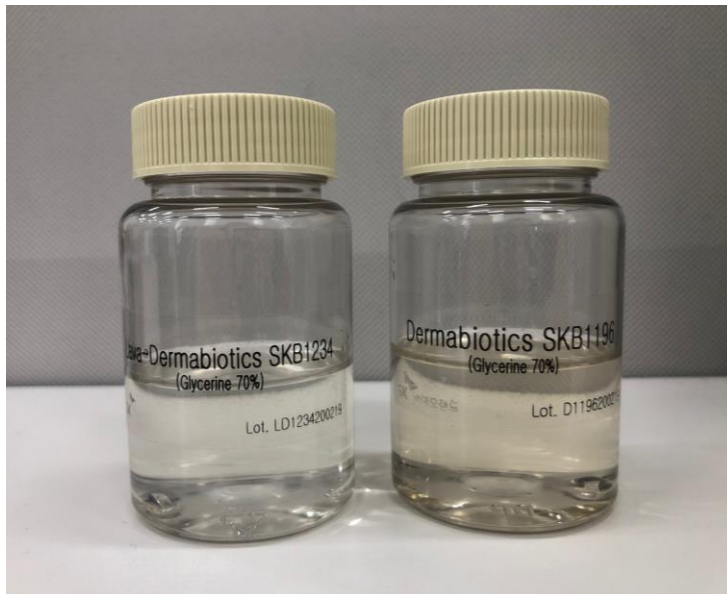


- ❖ Long-term (3 w)



Dermabiotics SKB1196

& Lava-Dermabiotics SKB1234



Transparent Probiotics Lysate Solutions **do not contain cell walls** which can cause skin toxicity and irritation.

Plus, it does not have turbidity, precipitate and odor specific to fermented product.

It contains only bio materials; cytosol of low molecular weight (2 kDa↓) which help penetrate into skin dermis and activate skin immune cells.

End of Document

